

**DEPARTMENT OF MATHEMATICS****Question Bank****Semester: I****Scheme: 2022****Subject: Mathematics-1 for CSE stream****Subject Code: BMATS101****Module-4**

| Q. No. | Question                                                                                                                                         | COs | BBT | Marks |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-------|
| 1      | Find the least positive values of $x$ such that<br>a) $71 \equiv x \pmod{8}$<br>b) $78 + x \equiv 3 \pmod{5}$<br>c) $89 \equiv (x + 3) \pmod{4}$ | CO4 | L2  | 6M    |
| 2      | Find the remainder when $135 \times 74 \times 48$ is divided by 7                                                                                | CO4 | L2  | 6M    |
| 3      | Find the remainder when $349 \times 74 \times 36$ is divided by 3                                                                                | CO4 | L2  | 6M    |
| 4      | Find the remainder when $175 \times 113 \times 53$ is divided by 11                                                                              | CO4 | L2  | 6M    |
| 5      | Find the unit digit in the remainder of<br>a) $7^{289}$<br>b) $7^{2013}$<br>c) $13^{37}$                                                         | CO4 | L2  | 6M    |
| 6      | Find the solutions of the linear congruence $11x \equiv 4 \pmod{25}$                                                                             | CO4 | L2  | 7M    |
| 7      | Solve the system of linear congruence $x \equiv 2 \pmod{3}$ ,<br>$x \equiv 3 \pmod{5}$ , $x \equiv 2 \pmod{7}$ by using CRT                      | CO4 | L3  | 7M    |
| 8      | Solve the system of linear congruence $x \equiv 3 \pmod{5}$ ,<br>$x \equiv 2 \pmod{6}$ , $x \equiv 4 \pmod{7}$ by using CRT                      | CO4 | L3  | 7M    |
| 9      | Solve: $2x + 6y \equiv 1 \pmod{7}$ , $4x + 3y \equiv 2 \pmod{7}$                                                                                 | CO4 | L3  | 6M    |
| 10     | Solve: $11x + 5y \equiv 7 \pmod{20}$ , $6x + 3y \equiv 8 \pmod{20}$                                                                              | CO4 | L3  | 6M    |
| 11     | Using Fermat's little theorem, find the remainder when $11^{104}$ is divided by 7                                                                | CO4 | L3  | 6M    |
| 12     | Using Fermat's little theorem, find the remainder when $2^{1000}$ is divided by 13                                                               | CO4 | L3  | 6M    |
| 13     | Find the remainder when $146!$ is divided by 149                                                                                                 | CO4 | L2  | 6M    |

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|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|----|
| 14 | Find the remainder when 14! is divided by 17                                                                                                                                                               | C04 | L2 | 6M |
| 15 | Find the rank of the matrix $\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 1 & 3 \\ 6 & 3 & 0 & -7 \end{bmatrix}$ .                                                                       | C05 | L2 | 6M |
| 16 | Evaluate the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 6 & 8 \\ 4 & 8 & 12 & 16 \\ 1 & 2 & 3 & 4 \end{bmatrix}$                                                                     | C05 | L3 | 6M |
| 17 | For what values of $\lambda$ and $\mu$ the system of equations<br>$2x + 3y + 5z = 9$ , $7x + 3y - 2z = 8$ , $2x + 3y + \lambda z = \mu$ has<br>(i) No solution<br>(ii) Unique solution                     | C05 | L3 | 6M |
| 18 | For what values of $\lambda$ and $\mu$ the system of equations<br>$x + y + z = 6$ , $x + 2y + 3z = 10$ , $x + 2 + \lambda z = \mu$ has<br>(iii) No solution<br>(iv) Unique solution<br>iii) Many solutions | C05 | L3 | 6M |
| 19 | Solve the following system of equations by Gauss elimination method.<br>$2x + y + 4z = 12$ , $4x + 11y - z = 33$ , $8x - 3y + 2z = 20$ .                                                                   | C05 | L3 | 6M |
| 20 | Solve the system of equations by Gauss-Jordan method.<br>$x + y + z = 9$ , $x - 2y + 3z = 8$ , $2x + y - z = 3$ .                                                                                          | C05 | L3 | 6M |
| 21 | Solve the system of equations by Gauss-Jordan method.<br>$2x + 5y + 7z = 52$ ; $2x + y - z = 0$ ; $x + y + z = 9$                                                                                          | C05 | L3 | 6M |
| 22 | Solve the following system of equations by Gauss Seidel method.<br>$10x + y + z = 12$ , $x + 10y + z = 12$ , $x + y + 10z = 12$ .                                                                          | C05 | L3 | 6M |
| 23 | Solve the following system of equations by Gauss Seidel method<br>$20x + 1y - 2z = 17$ ; $3x + 20y - z = -18$ ; $2x - 3y + 20z = 25$                                                                       | C05 | L3 | 6M |

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| 24 | Using Rayleigh's power method, find the dominant eigen value and the corresponding eigen vector of $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ by taking $[1 \ 0 \ 1]^T$ as the initial eigen vector [carryout 6 iterations].  | C05 | L3 | 7M |
| 25 | Using Rayleigh's power method, find the largest eigen value and the corresponding eigen vector of $A = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix}$ by taking $[1 \ 0 \ 1]^T$ as the initial eigen vector [carryout 6 iterations]. | C05 | L3 | 7M |